Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Fullstack AI engineer who **built AgentProbe end-to-end** - open-source LangGraph agent platform with plan-execute-react orchestration, 3-agent system (Orchestrator, RAG Tracer, Auto Fix), multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), RAG pipeline, 15-endpoint FastAPI backend, and React observability dashboard. Built agents. Shipped open-source tooling other engineers adopt. 8+ years production Python + full-stack delivery (JavaScript, React, PostgreSQL) across 120+ Agile sprints at enterprise scale. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LangChain / LangGraph | Agent Development & Orchestration | Open-Source Engineering | Full-Stack (React + FastAPI)

RAG + Vector Search | Multi-Provider LLM Routing | Production AI Systems | Prompt Engineering | LLM Evaluation Engine

AWS / Docker / CI/CD | Python (8+ years) | Cross-Functional Delivery

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built and delivered **20+ full-stack modules** (Python, JavaScript, TypeScript, PostgreSQL, React-equivalent) across **120+ Agile sprints** for **100K+ monthly active users** - end-to-end ownership from design to production deployment.
- Applied **Python, PyTorch, and OpenAI API** to automate HUD income limit data transformation, replacing manual Excel workflows across **1.6M records** - production AI applied to real data at scale.
- Led **Docker-based containerization** for 30%+ of modules; improved production query performance by **30%**; maintained **99.5% uptime** for 6+ months via TDD/BDD.
- Built **5+ internal debugging tools** adopted across multiple squads - reducing debugging time by **40%**. Open-source sensibility: build tooling that others adopt.
- Migrated client documents to AWS S3 resolving **90% of legacy storage constraints**; encrypted 100% of PHI via Libsodium - zero security incidents.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and databases; translated complex technical concepts - same discipline as making open-source tooling accessible to engineers with varying backgrounds.

PROJECTS

AgentProbe: Open-Source LangGraph Agent Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

End-to-end ownership of a production LangGraph agent platform - every architectural decision from greenfield to public open-source tool. **Agent orchestration**: 3-agent system (Orchestrator, RAG Tracer, Auto Fix) with plan-execute-react loop and custom tool calls. **Multi-provider routing**: Anthropic, OpenAI, Groq, Ollama - dynamic selection based on performance. **RAG + vector search**: ChromaDB, embedding pipeline, retrieval quality scoring. **Production eval engine**: 5-metric LLM-as-judge (factual accuracy, hallucination, retrieval, completeness, compliance) with regression tracking and CI/CD gates. **Full-stack**: 15-endpoint FastAPI backend + React observability dashboard. Shipped as open-source tooling adopted by other engineers. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to FHIR APIs via LLMs, BERT, and embeddings - full-stack Flask + React with HIPAA-compliant RBAC. **~60% retrieval time reduction.** Demonstrates applied AI end-to-end: model selection → pipeline architecture → production deployment. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → rigorous evaluation (MDI + Permutation Importance) → deployment → interactive stakeholder dashboard. **85%+ accuracy.** End-to-end applied ML - not just training but measurement, deployment, and communication.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Agent Orchestration, Plan-Execute-React, Tool Calls, Multi-Provider LLM Routing (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Prompt Engineering, LLM-as-Judge, LLM Evaluation

Full-Stack: React, FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MySQL, Docker

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ

Process: Open-Source Development, TDD/BDD, Integration Testing, Git, JIRA, Agile/Scrum